

Task 1: Install the Apache web server on your Linux machine using the package manager (apt or yum), and confirm the installation by running the command to check its version.

Ans :- Install Apache

```
sudo dnf install httpd -y
```

Verify Installation

```
httpd -v
```

Example Output

```
Server version: Apache/2.4.62 (CentOS Stream)
```

```
Server built: ...
```

Task 2. Start the Apache service using systemctl, then use systemctl status to verify it is running.

Ans :- Start Apache

```
sudo systemctl start httpd
```

Check Status

```
sudo systemctl status httpd
```

If Apache is running, you should see something similar to:

```
Active: active (running)
```

(Optional) Enable Apache at Boot

```
sudo systemctl enable httpd
```

**Task 3: Stop and restart the Apache service using systemctl, and observe the difference between stop, start, and restart commands.

Hint: Use 'systemctl stop apache2', 'systemctl start apache2', and 'systemctl restart apache2' (or 'httpd' on some systems).**

Ans :-

Stop Apache

```
sudo systemctl stop httpd
```

Verify

```
sudo systemctl status httpd
```

Expected:

Active: inactive (dead)

Start Apache Again

```
sudo systemctl start httpd
```

Restart Apache

```
sudo systemctl restart httpd
```

Difference Between the Commands

Command	Description
systemctl stop httpd	Stops the Apache service completely.
systemctl start httpd	Starts the service if it is stopped.
systemctl restart httpd	Stops the service and immediately starts it again. Commonly used after changing configuration files.

Task 4: Open your browser and access `http://localhost` or your Linux machine's IP address to verify that the Apache default web page is displayed.

Ans :-

Check Firewall (if enabled)

Allow HTTP traffic:

```
sudo firewall-cmd --permanent --add-service=http
```

```
sudo firewall-cmd --reload
```

Verify:

```
sudo firewall-cmd --list-services
```

You should see:

http

Find Your IP Address

ip addr

or

hostname -I

Example:

192.168.1.100

Open Browser

On the same machine:

http://localhost

or

http://127.0.0.1

From another computer:

http://192.168.1.100

You should see the **Apache HTTP Server Test Page** or the default Apache welcome page.

Task 5: Write a short comparison (3-4 lines) between Apache and Nginx web servers, mentioning one key use-case for each based on your research.

Ans :-

Apache is a process/thread-based web server that supports many modules and is easy to configure. It is commonly used for hosting PHP-based websites such as WordPress and applications requiring .htaccess support.

Nginx is an event-driven web server known for high performance and low memory usage. It is ideal for handling large numbers of concurrent connections and is commonly used as a reverse proxy, load balancer, and for serving static content.